



VATAS INFRASTRUCTURE

INTRODUCTION



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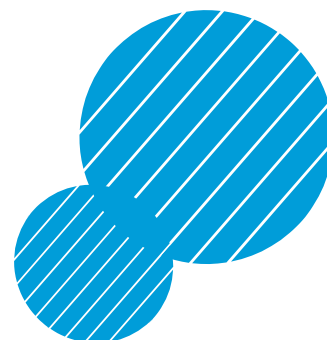


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Letter of Introduction

Date

To: Directors, Managers and Project Managers
Municipal and Regional Transportation / Infrastructure Divisions

Re: Introduction and Qualifications - Vatas Infrastructure Group Inc.

Vatas Infrastructure Group Inc. (Vatas) is pleased to submit this introduction and qualifications package for consideration on future Requests for Qualifications (RFQs) standing lists and assignment-based opportunities related to civil design, culverts and retaining walls, guiderail and roadside safety, municipal roadway improvements and construction phase services.

Vatas is Canadian owned, Ontario incorporated, P.Eng. led consulting firm focused on supporting municipalities and public agencies with the planning, design, and construction of their public works programs, including capital planning and asset management support for structures and roadway networks. Our team brings experience from York Region, the Ministry of Transportation of Ontario, the City of Toronto and the Toronto Transit Commission (TTC) through previous roles with public agencies and major consulting firms, now applied through a lean, senior-led practice.

We are particularly well suited to small and mid-scale municipal assignments, including:

- Municipal roadway and corridor improvements with active transportation elements;
- Minor capital works and resurfacing packages with associated civil works;
- Guiderail and roadside safety programs;
- Culvert repairs and other transportation structure rehabilitations.

Our team brings experience from comparable municipal programs. The attached booklet provides our company profile, team structure, representative project experience, and a summary of our delivery approach.

All services are delivered under Vatas' PEO Certificate of Authorization with professional liability insurance

We would welcome the opportunity to be considered for upcoming RFQs/RFPQs, rosters and assignment-based opportunities for municipal roadway projects and culvert/retaining wall rehabilitations and replacements. We are prepared to support your program either as prime consultant or as a specialist roads/structures subconsultant, delivering coordinated, constructable solutions.

Sincerely,



Karan Dev Sharma, P.Eng.
President & Principal Structural & Civil Engineer
Vatas Infrastructure Group Inc.

COMPANY

Overview

Who We Are

Vatas Infrastructure Group Inc. is an Ontario based, P.Eng. led consulting firm focused on municipal civil and structural transportation infrastructure. We are organized around the needs of municipalities and public agencies that must deliver reliable projects within constrained budgets and tight construction windows.

Vatas brings together experience gained on assignments for York Region, the Ministry of Transportation of Ontario, the City of Toronto, the TTC and major consulting firms, and applies it through a compact team with direct senior accountability. Senior-led delivery means senior engineers remain directly involved in scope, coordination, design decisions and deliverables—not only oversight.



Our Mission

To deliver fit for purpose designs and field support for municipal corridors, culverts and retaining walls that are safe, constructable and easy to tender, while integrating seamlessly with each client's capital program, standards and processes

Vision

To be a trusted technical partner to municipalities for small to mid-scale roadway, safety and structure assignments, known for practical solutions and dependable delivery.

Values

- **Public Service Mindset** - We recognize that every assignment ultimately serves residents, road users and taxpayers.
- **Technical Rigor, Practical Solutions** - We apply sound engineering while avoiding unnecessary complexity or cost.
- **Accountability** - Senior engineers stay close to the work, decisions are documented and commitments are met.
- **Collaboration** - We work as an extension of municipal staff, coordinating effectively with internal departments, utilities and agencies

CORE SERVICES

Vatas is intentionally focused on a narrow but deep set of core services

Municipal Roadway & Corridor Design

Resurfacing and minor capital works, localized widening, intersection and corridor improvements, drainage adjustments, active transportation elements and roadside safety / guiderail design, with staging and utility coordination integrated from the outset.



Vatas Advantage

Best suited to small and mid-scale municipal assignments such as localized geometry and safety improvements, guiderail and Active Transportation upgrades, and resurfacing packages where senior engineers stay directly involved in design and staging decisions.

Culverts & Retaining Walls

Planning level options analysis (SPR/SDR support), rehabilitation and replacement design, end treatments, staged construction strategies and tender-ready drawings, specs and quantities for culverts and retaining walls on municipal roads.



Vatas Advantage

Focused on culvert repairs and small structure rehabilitations and providing cost-effective, state-of-the-art, constructable solutions that balance cost, risk, traffic impacts, and life-cycle performance.

Construction Services & Field Inspection

Site inspection and technical support to contract administrators, including RFIs, shop drawing reviews, quantity verification, deficiency tracking, payment certificate support, and change order / change evaluation input on roadway, safety and culvert-related contracts.



Vatas Advantage

Senior-led inspectors who understand design intent and provincial/municipal standards, provide clear documentation and advice so contracts can be administered confidently on schedule and within budget.

Digital Support for Programs

Straightforward, maintainable tools to track locations, quantities and status across guiderail, resurfacing and culvert programs, improving visibility for staff without imposing complex new systems.

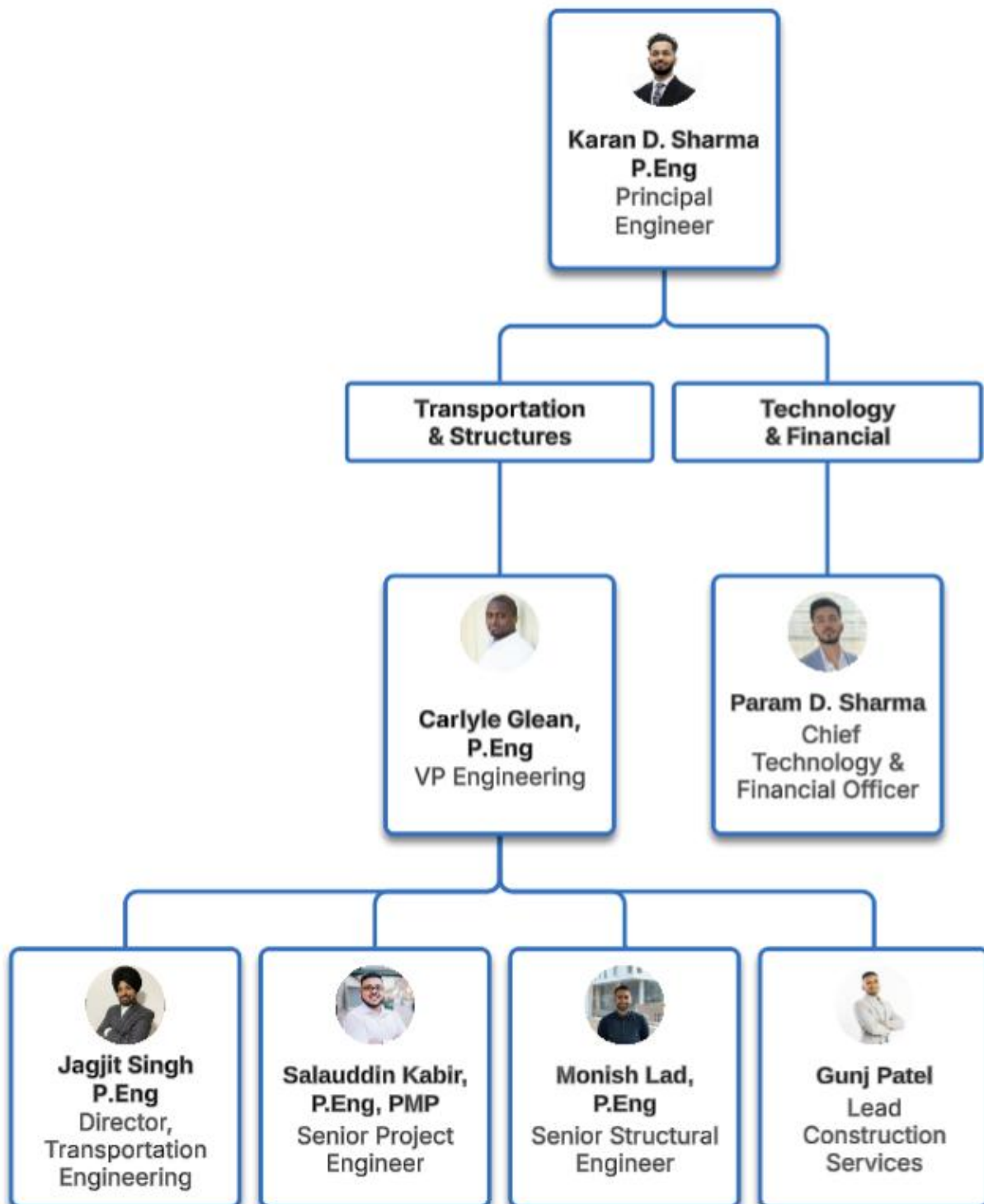


Vatas Advantage

Lightweight dashboards and trackers aligned with municipal capital and asset management processes, helping staff manage multi location programs efficiently with clear, at a glance status across their network.

ORGANIZATIONAL STRUCTURE & KEY PERSONNEL

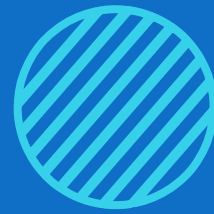
Organizational Structure



Meet our Experts

Vatas is led by a senior team of Professional Engineers and inspectors focused on municipal roads, transportation structures, transit corridors and construction support

SENIOR-LED MUNICIPAL ENGINEERING. PRACTICAL, TENDER-READY DELIVERABLES



Karan D. Sharma, P.Eng President, Principal Engineer

Karan is a Project Manager with experience on municipal structure and roadway programs, including multistructure renewals and minor capital works with 5 years' experience. Karan founded Vatas Infrastructure Group with the vision of helping develop municipal infrastructure.



Param D. Sharma Chief Technology Officer

Param is a full-stack developer focused on building program tracking and reporting tools for guiderail, resurfacing and culvert packages.

Quality Over Quantity

Our lean, senior-led structure means the experts you see here are the ones who plan, design and support delivery of your projects, ensuring consistent attention and technical rigour on every assignment.



Carlyle Glean, P.Eng

Vice President, Engineering

Senior municipal transportation engineer with 18+ years' experience in geometric design of highways and municipal roadways, roadside safety, drainage/stormwater management and corridor design



Jagjit Singh, P.Eng

Director, Transportation Engineering

Transportation engineer with 18+ years' experience in road and highway design, LRT corridor work, transit, drainage and utilities coordination



Saluddin Kabir, P.Eng, PMP

Senior Project Engineer, Transit & Municipal

Project engineer with experience across design, construction and project management on TTC station works, LRT/BRT programs and municipal transportation projects



Monish Lad, P.Eng

Senior Structural Engineer, Bridges and Culverts

Structural engineer focused on bridge and culvert rehabilitation and replacement, including liners, box culverts and retaining walls.



Gunj Patel

Lead, Construction Services & Field Inspection

Site inspector and construction services lead with experience on roadways, watermain, roofing, and sewer works, supporting contract administration in the field.

KEY STAFF PROFILES

Karan Dev Sharma, P.Eng.

President & Principal Civil/Structural Engineer

Karan Dev Sharma is a Professional Engineer and project manager with 5+ years of experience supporting municipal bridge, culvert, retaining wall and roadway programs for public sector clients. His background includes structural renewal coordination, planning-level evaluation and options development (SPR/SDR style support), multi-discipline coordination, and construction-phase technical support. He leads Vatas' technical delivery and client coordination, ensuring assignments are executed with clear accountability and fit-for-purpose outcomes.

Practice focus

- Bridge, culvert and small structure rehabilitation and replacement
- Municipal roadway rehabilitation, minor capital and resurfacing programs
- Structural planning/evaluation and options development
- Capital planning and coordination for municipal structures portfolios

Param Sharma

Chief Technology & Financial Officer

Param Sharma is a software and cloud engineering professional with experience developing data driven platforms, APIs and automation tools in enterprise environments. At Vatas, he enables program visibility and efficient delivery by building practical dashboards and trackers for multi-site municipal assignments, including guiderail, resurfacing and culvert packages. He also supports financial modelling and analysis for multi-year capital work to help teams plan and report consistently.

Practice focus

- Program dashboards and tracking tools for municipal infrastructure
- Data structuring and handover aligned with client asset/capital systems
- Financial modelling, forecasting and scenario analysis for capital programs
- Internal cloud, security and data management practices

Carlyle Glean, P.Eng.

Vice President, Engineering

Carlyle Glean is a senior municipal transportation engineer with 18+ years of experience in planning and designing highways, municipal roadways and corridors across Ontario. His portfolio includes rehabilitation, widening and new corridor projects, with extensive involvement in roadway geometry, grading, drainage/stormwater management, erosion and sediment control, roadside safety, utility coordination, traffic management and contract documentation. He provides senior technical oversight and review to ensure roadway deliverables are coordinated, constructable and tender ready.

Practice focus

- Municipal and highway corridor design (rehabilitation, widening and greenfield)
- Roadway geometry, grading, drainage and roadside safety design
- Traffic management and staging for construction of road projects
- Multi-disciplinary coordination and delivery of tender ready roadway contracts

Jagjit Singh, P.Eng.

Director, Corridor Design & Transportation Engineering

Jagjit Singh is a transportation engineer with 18+ years of experience delivering urban and rural road, highway and transit corridor design in Canada and internationally. His experience spans alignment and intersection layout, corridor grading and drainage, transit integration and utilities coordination, with a strong focus on producing complete design packages for municipal capital works. He leads day-to-day corridor design delivery, coordinating technical interfaces and preparing drawings and quantities suited to resurfacing and minor capital assignments.

Practice focus

- Urban and rural roadway and corridor design
- Transit adjacent road and intersection design
- Corridor drainage, grading and utilities integration
- Design documentation for municipal road and streetscape projects

Ahmed Salauddin (Sal) Kabir, P.Eng., PMP

Senior Project Engineer, Transit & Municipal Transportation

Sal Kabir is a civil engineer and PMP with experience across major transit and municipal transportation projects in the GTHA. His work includes technical and coordination roles on BRT and LRT corridors, TTC station and accessibility projects, municipal road programs and development related transportation improvements. He supports complex corridor delivery by strengthening stakeholder coordination, staging integration, and project controls, bridging design intent with construction and owner requirements.

Practice focus

- Transit adjacent municipal road, intersection and corridor projects
- Technical advisory and coordination on capital transportation programs
- Construction phase coordination and contract administration support
- Multi-stakeholder urban corridor assignments involving transit agencies and municipalities

Monish Lad, P.Eng. ,M.A.Sc

Senior Structural Engineer, Bridges & Culverts

Monish Lad is a structural engineer specializing in bridge and culvert rehabilitations and replacements, including box culverts, short span structures, culvert lining, and full replacements. His experience includes structural design reports, evaluations, detailed design and contract documentation for municipal clients. He supports staged construction planning and provides structural QA/QC to ensure structure-related deliverables are technically sound, coordinated and suitable for tender.

Practice focus

- Rehabilitation and replacement design for bridges and culverts
- Structural planning/evaluation and options analysis for small structures
- Retaining walls and related structural components for municipal roads
- Emphasis on utilizing modern, state-of-the-art repair products and methodologies to maximize durability and cost efficiency
- Staged construction strategies and input to tender drawings and estimates

Gunj Patel, B.Eng., EIT

Lead, Construction Services & Field Inspection

Gunj Patel is a construction inspector and civil engineer with field experience on municipal capital works and private development projects, including watermain, storm and sanitary sewers, roadway reconstruction and related civil works. He supports contract administration through daily full-time site inspections, documentation, payment certificate validation, contractor evaluations, field testing coordination, and deficiency management, helping ensure work conforms with design intent, specifications and municipal standards.

Practice focus

- Field inspection for roadway and underground infrastructure construction
- Monitoring workmanship, testing and compliance with contract requirements
- Site reporting, deficiency tracking and input to payment certificates
- Track and validate extra work orders.
- Liaison with contractors, residents and municipal staff during construction
- Liaison with municipal operations for shutdowns, closures, and notice distribution to residents and businesses.
- Assist design teams with as-built preparation.

SERVICES & CAPABILITIES

Vatas provides focused municipal civil and structural services covering culverts and retaining walls, municipal roadway and active transportation design, guiderail and roadside safety, construction support, and digital tools for program tracking.

STRUCTURAL ENGINEERING - CULVERTS & RETAINING WALLS

Condition assessment and planning support	for culverts and retaining walls, including input to structural rehabilitation options analysis.
Rehabilitation and replacement strategies	for culverts (lining, partial reconstruction, full replacement) and walls (refacing, localized repairs, full replacements)
Detailed design and contract documents	for culvert barrels, headwalls, wingwalls, retaining walls and related appurtenances.
Staged solutions	Coordinate with roadway and active transportation works to maintain services where required.
Quantities and opinion of probable construction cost (OPCC)	for both stand alone and bundled programs.

MUNICIPAL ROADWAY & HIGHWAY DESIGN

Minor capital & resurfacing design	Pavement rehabilitation, localized widening and safety related civil works for municipal and regional programs.
Streetscape & accessibility upgrades	Curb, sidewalk and boulevard improvements including drainage adjustments and accessibility upgrades.

Active transportation integration	Cycling and pedestrian elements including bike lanes, multi-use paths, crossings and transit connections.
Guardrail & roadside safety design	Barrier and roadside safety design integrated with geometry, slopes, culvert ends and retaining walls..
Intersections & corridor improvements	Turn lanes and localized alignment refinements for arterials and collectors.
Tender-ready design documentation	Plans/profiles, typical sections, grading, drainage and contract drawings suitable for tender.

TRANSIT & CORRIDOR ADVISORY

Transit corridor roadway modifications	Road and intersection modifications supporting stations, stops, bus operations and corridor requirements.
Complete streets & Active Transportation coordination	Integration of active transportation and complete streets elements within transit influenced corridors.
Utility & property interface coordination	Coordination of roadway design with transit infrastructure, utilities and property adjustments.
Staging & detour input	Municipal road and AT perspective on staging, detours and maintaining access during construction.
Third party design review	Reviews for impacts to municipal roads, culverts, retaining walls and roadside safety

TRAFFIC MANAGEMENT, STAGING & DETOURS

Early stage staging concepts	High-level staging concepts for planning documents and early design phases.
Traffic management input to design	Lane configurations, temporary alignments and work zone layouts reflected in contract documents.
Active Transportation and transit detour continuity	Temporary cycling and pedestrian routes and transit detour considerations during construction.
Structures + roadway sequencing	Staging integration for culvert and retaining wall works to support feasible construction sequencing.
Stakeholder coordination support	Coordination input with internal groups, emergency services, transit and adjacent stakeholders.

CONSTRUCTION SERVICES & FIELD INSPECTION

Site inspection services	Inspection for road resurfacing, linear infrastructure, commercial roofing, culvert works, guiderail installations and Active Transportation elements.
Testing and compliance monitoring	Monitoring workmanship and testing (asphalt, concrete, compaction) to contract requirements. Monitor in-situ testing performed by geotechnical technicians.
Documentation & deficiency tracking	Daily reports, photo logs and deficiency tracking aligned with municipal standards. Detailed tracking logs for change order works and payment certificate issuance.
Contract administration support	Support for RFIs, shop drawing review comments and change evaluation input. Assistance for design team to prepare as-built drawings.

Quantity verification & payments

Quantity verification and input to payment certificates based on tender items and completed work.

DIGITAL & DATA SOLUTIONS FOR INFRASTRUCTURE

Program dashboards & trackers

Track locations, quantities, budget and progress for guiderail, resurfacing, culvert and AT packages.

Standardized data structures

Design output data structured for easier import into municipal asset and capital planning systems.

Reporting for management and Council

Simple status, risk and milestone reporting to support internal management and Council reporting.

Automation for consistency

Automate repetitive checks and summaries (e.g., quantity rollups) to improve consistency and efficiency.

Representative Experience of Vatas Personnel

The projects summarized in this section reflect the individual experience of Vatas personnel gained while employed at previous public agencies and consulting firms. Similar services would be delivered under Vatas Infrastructure Group Inc. on future assignments.

Summary of Relevant Experience - Team Overview Matrix

Name / Role	Years of Experience	Municipal Roadway & Corridor / AT & Guiderail	Culverts, Retaining Walls & Small Structures	Transit / Rapid Transit & Corridor Advisory	Construction Administration & Field Inspection	Typical Role on Assignments
Karan Dev Sharma, P.Eng., PMP – President & Principal Civil/Structural Engineer	5+	Capital road programs, local and arterial roadway design, intersection improvements, and active transportation elements (sidewalks, multi-use paths, crossings).	Program level delivery of culvert and small structure rehab/replacement under a regional structural renewal program.	Support to LRT and subway expansion projects with focus on road/traffic staging and municipal interfaces.	Design/field coordination for municipal roads and structures, including support during construction.	Project Director; Project Manager; structural/roadway planning and coordination lead.
Param Sharma – Chief Technology & Financial Officer	3–5 (software & cloud)	Provides data and dashboard support for road and AT programs (tracking locations, budgets, schedules).	Not applicable (non-design role; supports program data and reporting).	Supports enterprise initiatives with digital tooling, workflow automation, and reporting integration.	Provides tools for cost tracking, progress dashboards and documentation management.	Chief Technology & Financial Officer; Lead for digital and data solutions; supports bids, analytics and reporting.
Carlyle Glean, P.Eng. – Vice President, Municipal Transportation Design	17	Extensive experience on arterial rehabilitation, widenings, complete streets and safety/guiderail upgrades for municipalities and regions.	Road projects involving culverts, slopes and retaining walls are integrated with roadway and safety works.	Corridor/highway interface experience where municipal roads interact with provincial and transit corridors.	Leads multi-disciplinary teams delivering road contracts that include CA/inspection components.	Senior Project Manager; Municipal roadway and corridor design lead; QA/QC for roads/AT/guiderail.

Jagjit Singh, P.Eng. – Director, Corridor Design & Transportation Engineering	16	Urban and rural road and corridor design, including roundabouts, intersections, and corridor improvements.	Culvert and small structure interfaces within roadway and corridor assignments.	Major transit corridors (LRT/BRT, subway extension) – roadway design, staging and corridor integration.	Coordinates constructability and staging inputs into design packages.	Corridor design lead; senior transportation engineer; staging and AT/safety design specialist.
Ahmed Salauddin (Sal) Kabir, P.Eng., PMP – Senior Project Engineer, Transit & Municipal Transportation	6+	Capital road portfolios, intersection upgrades, and state of good repair roadway programs.	Road/structure interfaces within municipal capital works programs.	BRT/LRT technical advisory, P3 support (procurement documents, technical reviews, staging and property).	Background in inspection/CA for TTC and municipal works.	Senior project engineer; technical advisor on roads/transit; coordination and change management.
Monish Lad, P.Eng. – Senior Structural Engineer, Culverts & Retaining Walls	5+	Provides structural input for roadway programs that include culvert and retaining wall components.	Specialist in culvert rehabilitation/replacement and retaining wall/MSE wall design for regional and county roads.	Supports assignments where culvert/retaining wall works interface with transit or highway corridors.	Supports permitting and field coordination for structure works.	Senior structural engineer; culvert and retaining wall design lead; contributor to planning/SDR/SPR.
Gunj Patel – Lead, Construction Services & Field Inspection	5+	Field inspection on roadway and utility works within corridors, including coordination with residents and transit.	Inspection of works affecting site services and small structures		support contract administrator with detailed logs and reports of daily site activities, review and issue CD's, RFI's, RFQ's, Payment Certificates and Contractor Evaluations.	

Municipal Roadway & Corridor Projects (Team Experience)

Project Name & Location	Client / Owner	Capital Cost (Approx.)	Project Type / Scope	Delivery Model	Role of Vatas Personnel	Key Responsibilities (Summary)	Relevant Vatas Team Member(s)	Year(s)
Williams Parkway Complete Streets, McLaughlin Rd to Dixie Rd, Brampton	City of Brampton	≈ \$30M	Reconstruction of ~5.7 km arterial corridor including lane narrowing for traffic calming, new multi-use paths on both sides, intersection and signal upgrades, illumination, landscaping, drainage improvements and rehabilitation of multiple structures.	Design-Bid-Build	Transportation Design Lead; Senior Transportation Engineer	Corridor design, active transportation and roadway geometry, staging approach, utility conflict identification, coordination with regional, municipal and conservation stakeholders; support tender documentation.	Carlyle Glean; Jagjit Singh	2022–2023
Derry Road & Bramalea / Menkes / Alstep Improvements (Bombardier Off-Site Works), Mississauga	Region of Peel / Bombardier Aerospace	≈ \$20M roads (within larger program)	Widening and intersection upgrades on Derry and Bramalea Roads, Menkes Drive widening and new Alstep Drive extension to a signalized intersection; active transportation, boulevard and drainage improvements, integration with adjacent industrial and airport uses.	Class EA + Detailed Design (DBB)	Project Manager / Project Engineer; Roadway Designer	Options development, horizontal/vertical alignment, AT and access design, swept path analysis, staging and property impact assessment, coordination with Region, City, airport and private stakeholders; preparation of design drawings and estimates.	Carlyle Glean; Karan Dev Sharma	2020–2023
King Street South & Scott Street Rehabilitation, Caledon	Town of Caledon	≈ \$0.4M	Local road rehabilitation including pavement reconstruction, slope stabilization, drainage improvements and installation of cable guiderail for roadside safety.	DBB	Project Manager / Project Engineer	Detailed road design, slope stabilization measures, drainage upgrades, guiderail layout, and coordination of permits with conservation and regulatory bodies; preparation of tender-ready documents.	Carlyle Glean	2020

Highway 7 & Sixth Line Intersection Improvements and Sixth Line Reconstruction, Guelph-Eramosa	Township / Contractor / MTO interface	≈ \$5.7M	Intersection capacity and safety improvements at Highway 7, highway widening for turn lanes, guiderail replacement, utility relocation, and reconstruction of Sixth Line including profile, shoulders and safety upgrades.	DBB	Project Manager / Project Engineer	Intersection and corridor design, guiderail and roadside safety upgrades, coordination of utilities and property impacts, approvals with MTO and the municipality, tender drawings and quantities.	Carlyle Glean	2021–2023
Major & Minor Capital Road Design Portfolio, Regional Network	Regional Municipality	≈ \$100M program (multi-year)	Portfolio of road reconstruction and intersection improvement projects delivering state of good repair upgrades, safety improvements, staging strategies and contract documents across multiple locations.	Program of DBB assignments	Technical Advisor; Internal Program Coordinator	Technical review of designs, staging and safety reviews, stakeholder coordination (municipalities, transit agencies, provincial partners), input to tender and contract administration requirements, program-level tracking and reporting.	Ahmed S. Kabir; Karan Dev Sharma	2023–2025
O'Brien Avenue Roadway Design, Stouffville	Town of Whitchurch–Stouffville	– (local capital project)	Urban roadway design including horizontal/vertical alignment, cross-sections and integrated storm, sanitary and watermain networks, with connections to local active transportation facilities.	DBB – Detailed Design	Roadway Designer	Full Civil 3D corridor modelling, underground servicing design, grading and cross-section development, coordination with municipal standards and staff comments.	Karan Dev Sharma	2021–2022
County Road 53 & Carson/Seadon Road Roundabout, Springwater	County of Simcoe	Intersection capital project	Class EA and detailed design for a rural single lane roundabout including roadway approaches, utility relocations, property impacts, drainage features and contract documents.	EA + DBB	Senior Transportation Engineer	Roundabout geometry and modelling, design drawings, utility conflict and property impact plans, support to tender documentation and staging concepts.	Jagjit Singh	2019–2022

Culvert & Retaining Wall Projects (Team Experience)

Project Name & Location	Client / Owner	Capital Cost (Approx .)	Structure Type & Scope	Delivery Model	Role of Vatas Personnel	Key Responsibilities (Summary)	Relevant Vatas Team Member(s)	Year(s)
Rouge River Culvert Replacement, Highway 7	Regional Municipality	≈ \$10.2M	Replacement of a long span corrugated steel pipe culvert with a precast concrete box culvert, including dewatering via bypass pipe, staged excavation under a regional arterial, utility relocations and environmental permitting.	DBB	Lead Structural Engineer	Structural planning and preliminary design, cost estimating, detailed design of culvert and appurtenances, preparation of contract drawings, coordination with utilities and permitting agencies.	Monish Lad; Karan Dev Sharma (program oversight)	2024-Present
Keele Street Culvert Rehabilitation	Regional Municipality	≈ \$1.6M	Rehabilitation of twin open footing concrete culverts using FRP liners and associated retaining walls, delivered under staged construction.	DBB	Structural Engineer	Structural evaluation, development and detailing of FRP lining solution, design of retaining walls, preparation of tender documents, support to environmental and regulatory submissions.	Monish Lad; Karan Dev Sharma	2024-Present
Yonge Street Culvert Rehabilitation	Regional Municipality	≈ \$1.7M	Rehabilitation of an existing concrete box culvert using HDPE lining and MSE retaining walls, with staged construction under a regional road.	DBB	Structural Engineer	Preliminary design, cost estimating, detailed design of lining and wall systems, coordination with conservation authority and federal fisheries approval requirements.	Monish Lad; Karan Dev Sharma	2023-2024
King Road Culvert Rehabilitation	Regional Municipality	≈ \$3.4M	Rehabilitation of CSP culvert using FRP liner, plus new tunneled overflow pipe and MSE retaining walls; micro tunnelling under roadway with complex geometry and	DBB	Structural Engineer	Development of rehabilitation alternatives, micro tunnel alignment and overflow pipe configuration, retaining wall design, contract	Monish Lad	2023-Present

			constraints.			drawings and coordination with regulatory authorities.		
County Road 17 Culvert Replacements	County	≈ \$2.8M	Replacement of multiple CSP culverts with precast concrete box culverts delivered under a single multi-site contract with staged construction.	DBB	Structural Engineer / Proposal Lead	Preliminary design reports, quantities and cost estimates, detailed design of replacement culverts, design drawing preparation, multi-site staging strategies and contract documentation.	Monish Lad	2024-Present
Structural Renewal Program - Multistructure Portfolio	Regional Municipality	≈ \$22M per year	Annual program of bridge, culvert and retaining wall rehabilitation and replacement projects across a regional road network.	Program of DBB assignments	Planning & Design Coordinator	Program management support, prioritization and scoping, coordination of consultants, design reviews, procurement support, schedule and budget tracking.	Karan Dev Sharma	2023-Present

Transit & LRT/BRT / Corridor Advisory Projects (Team Experience)

Project Name & Location	Client / Owner	Capital Cost (Approx.)	Project Type / Scope	Delivery Model	Role of Vatas Personnel	Key Responsibilities (Summary)	Relevant Vatas Team Member(s)	Year(s)
Hurontario Light Rail Transit (HuLRT), Port Credit to Brampton	Metrolinx / Infrastructure Ontario with municipal partners	≈ \$4.6B	18 km LRT corridor through urban and suburban environments; roadway and track interface design, structures interfaces, constrained cross-sections and staging along major arterials.	DBFOM P3	Senior Transportation Engineer; Design / Coordination Support	Roadway alignment and cross-sections, staging concepts, integration of LRT with existing roads review of designs and coordination with agencies.	Jagjit Singh; Karan Dev Sharma	2019-Present
Eglinton Crosstown West Extension (ECWE) - Advance Tunnel	Metrolinx / Infrastructure Ontario	≈ \$700M+ (tunnel package)	Twin bored tunnel with launch and extraction shafts, headwalls and cross-passages forming part of a subway extension; roadway and traffic interfaces at shaft sites.	DBFOM P3 (Design Support)	Project Engineer - Roads; Design Coordination	Road and traffic management input, access and staging considerations around shaft sites, coordination of tunnel interfaces with surface roads and utilities.	Jagjit Singh; Karan Dev Sharma	2021-2022
Dundas Bus Rapid Transit (BRT), Mississauga Corridor Advisory	Regional / Municipal Client	Major BRT program	Reference concept design and technical advisory for BRT corridor including roadway, utilities, structures, property and grading impacts along Dundas Street.	P3 / DBFOM - Technical Advisor	Technical Advisor	Review and refinement of concept designs, grading and profile review, impact and conflict identification, support to procurement documentation.	Ahmed S. Kabir	2022-2023
TTC Rail Expansion - Capital Projects	Toronto Transit Commission	Multi-project capital program	Planning and design support for advanced rail expansion projects including stations and corridor elements.	Program Assignments	Senior Project Engineer	Coordination of design phases, stakeholder input, schedule and cost tracking and support to technical deliverables for TTC capital works.	Ahmed S. Kabir	2025-Present

Construction Administration & Field Inspection Projects

Project Name & Location	Client / Owner	Facility / Asset Type	Capital Cost (Approx.)	Role of Vatas Personnel	Key Responsibilities (Summary)	Relevant Vatas Team Member	Year(s)
Watermain Replacement on Williams Parkway (HDD), Brampton	Region of Peel	HDPE Watermain Installation	\$5.7 Million	Construction Inspector	Full-time inspection of HDD installation and associated works, daily reporting, coordination of QA testing, support to payment certification, liaison with residents and transit for road impacts and closures. Liaison with municipal operations and inspectors for shutdowns, pressure testing, chlorination, and commissioning.	Gunj Patel	2022-2024
Airside Sanitary Sewer Installation, Pearson International Airport	Bombardier Aviation / GTAA	Airside sanitary sewer infrastructure	\$2.3 Million	Construction Inspector	Inspection of sanitary pipe installation, excavation, and backfilling under airside constraints and requirements; coordination with airport operations, QA documentation, support to progress meetings and payments. Dispute resolution between client and contractor.	Gunj Patel	2021-2022
Tanks & Site Services - Aerospace Facility	Bombardier Aviation	Underground tanks and site services	\$19 Million	Construction Inspector	Inspection of underground tank installations (storm, foam, firewater) and associated site services; coordination with multiple trades and testing agencies; daily records and issue tracking.	Gunj Patel	2021-2022
Commissioners Transfer Station & Ingram Yard Roof Replacement, Toronto	City of Toronto	Roof Replacement at Solid waste facility and yard buildings	\$2.2 Million	Construction Inspector	Inspection of roof replacement and building envelope works; daily reports; review of submittals and payment applications; coordination with facility operations staff. Review of RFI's and track CD works.	Gunj Patel	2024-2025
Watermain Replacement on Dundas St East (CPP Watermain), Mississauga	Region of Peel	CPP Watermain	\$12.4 Million	Construction Inspector	Full-time inspection of CPP watermain installation and associated works, daily reporting, coordination of QA testing, support to payment certification, liaison with business owners for shutdowns and connections. Liaison with municipal operations and inspectors for shutdowns, pressure testing, chlorination, and commissioning.	Gunj Patel	2025-present

PROJECT DELIVERY APPROACH

Vatas applies a **disciplined, scalable project delivery approach** across all assignments: define the problem clearly, evaluate options, design for constructability and safety, and support municipalities through tender and construction. The same logic scales from a single culvert to a multi-corridor program.

Approach to Municipal Roadway & Corridor Assignments (including Active Transportation & Guiderail Design)

For roadway work, we treat each corridor as a whole system: pavement, drainage, active transportation, transit operations and roadside safety.

Our typical approach:

Define outcomes early – Clarify whether the priority is state of good repair, safety, capacity, AT, transit operations, or a mix, and set design criteria accordingly

Map constraints properly – Use survey, traffic/collision data, utilities, property, drainage and existing AT/transit routes to avoid late surprises.

Develop clear options – Prepare a focused set of cross-section and intersection alternatives (lanes, AT, turn lanes, guiderail) and compare them on safety, operations, cost and constructability.

Design for construction, not just for drawings – Refine alignment, grading, drainage, AT and **guiderail/roadside safety** as one coordinated package, with staging and traffic management considered from the outset.

Support tender and delivery – Produce quantities and OPCCs in municipal friendly formats, respond to tender queries, and provide concise design clarifications during construction

Result: **fit-for-purpose corridor designs** that are easier for municipalities to tender, build and maintain.

Approach to Culvert & Retaining Wall Rehabilitation and Replacement

For culverts and retaining walls, our focus is on choosing the **right level of intervention**—no over-design, no unmanaged risk.

Our typical approach:

Start from condition and risk – Review inspections, hydraulic models, geotechnical results and constraints (utilities, environmental, access) and agree on the real drivers with staff.

Compare realistic strategies – Evaluate lining, partial or full replacement, and wall refacing/reconstruction across the program, considering lifecycle, construction risk, traffic impacts and cost. Vatas applies practical, durable and cost-effective rehabilitation solutions.

Advance the preferred option efficiently – Lock in configuration (span, alignment, wall type, end treatments), rehabilitation strategy and methodology, and confirm hydraulic and structural performance requirements.

Integrate staging and traffic into the design – Build staging, traffic control, erosion/sediment control and utility protection into the detailed design, not as an afterthought.

Deliver complete, coordinated tender packages – Provide structural design, drawings, specifications, quantities and OPCCs aligned with municipal contract structures, and stay involved through RFIs and shop drawing reviews.

Result: **defendable rehabilitation/replacement decisions** and construction packages that reduce change order risk.

Approach to Transit & Corridor Advisory Assignments

On transit projects, Vatas' role is to protect and enhance the **municipal road and corridor interests** within larger, multi-party undertakings.
Our typical approach:

Clarify the municipal perspective – Identify how the LRT/BRT/subway affects lanes, intersections, AT continuity, access, property and utilities.

Review concepts through a road/AT lens – Assess alignments, stops/stations and concept designs for their implications on operations, safety, AT and local access.

Shape clear, road focused requirements – Help municipalities define corridor requirements in output specs and reference concepts so proponents must address them.

Support technical reviews – Provide targeted, structured comments on proponent designs, focusing on municipal operations, safety and maintainability.

Advise on staging and detours – Input on road, AT and access staging during transit construction, helping staff communicate realistic impacts to the public.

Result: the municipality **keeps control of its corridor interests** even when delivery is through P3s or multi-agency teams.

Coordination with Municipal Staff, Utilities and Stakeholders

Because Vatas is senior-led, clients work directly with decision makers. Coordination is planned and documented, but not bureaucratic.
Our typical approach:

Set up a clear coordination plan – Identify core and extended municipal team members and agree on meeting cadence and decision points.

Use an issues/decisions log – Track key questions, inputs and resolutions so nothing is lost between checkpoints.

Engage utilities early and cleanly – Circulate consolidated base plans, prepare composite utility drawings and maintain a conflict/resolution log.

Align with regulators – Confirm permits and timelines at the start and provide focused technical submissions that directly answer their requirements.

Support public and stakeholder communication – Prepare clear figures and concise summaries to help staff at PICs, on project webpages and in Council/Committee reports.

Result: fewer surprises at 90% design or during tender, and a clear record of “who decided what and why.”

Quality Assurance / Quality Control

Our QA/QC approach is **structured and proportionate to project scale**—rigorous enough for municipal standards, efficient enough for small assignments and program work. Our typical approach:

Plan QA/QC at project start – Define stage gates (e.g., 30%, 60%, 90%, pre-tender) and assign reviewers by discipline.

Apply structured discipline checks – Use checklists for geometry, AT, drainage, culverts/walls, roadside safety and staging, with independent reviewers for key elements.

Run coordination reviews – Conduct combined plan/profile and typical section reviews to catch interactions between road, AT, drainage, structures, utilities and traffic control.

Control versions and records – Maintain consistent file naming, revision histories and marked review sets to avoid confusion.

Feed lessons back into future work – Capture lessons learned and client feedback and update templates, details and checklists accordingly.

Result: coordinated, constructable design packages that municipalities can rely on at tender and in the field, without unnecessary process overhead.

CONSTRUCTION PHASE & FIELD SERVICES

Vatas supports municipalities beyond design by providing **practical, field focused services** during construction. Our goal is to help contract administrators deliver projects safely, on schedule and within budget, with clear, documented decisions when conditions change.

Role of Vatas During Construction

Vatas can support the construction phase in two main ways:

- **Design Engineer of Record Support**
 - Respond to RFIs related to roadway, active transportation, culvert and retaining wall design.
 - Review shop drawings, temporary works and alternative products where they affect design intent, safety or durability.
 - Provide rapid, documented design input for field changes driven by site conditions, utilities or staging constraints.
- **Field Support to Contract Administration Teams**
 - Attend pre-construction and progress meetings as required to clarify design requirements and constraints.
 - Assist contract administrators in evaluating proposed changes, extensions of time and cost implications from a design perspective.
 - Prepare Payment Certificates and Contractor Evaluations.
 - Support municipalities in interpreting permit conditions (e.g., in-water work windows, erosion control) as they apply on site.

Result: **continuity between design and construction**, with the same senior team members who developed the design available to help resolve field issues efficiently.

Inspection, Contract Administration and Site Coordination

Where municipalities require site-based support, Vatas can provide **experienced inspection and coordination staff** to work under the direction of the owner's contract administrator.

Typical responsibilities include:

- **Construction Inspection**
 - Full-time or part-time inspection of roadway, culvert, retaining wall, utility and associated works.
 - Daily reports documenting progress, workmanship, testing, weather, quantities and notable site events.
 - Verification that materials and construction methods conform to contract documents and applicable standards.
 - Inspect pressure testing, chlorination, in-situ compaction and concrete testing.
- **Support to Contract Administration**
 - Assist with measurement and verification of quantities for payment certificates.

- Track non-conformances, deficiencies and corrective actions.
- Track and analyze work performed under Change Directives.
- Perform Contractor Evaluations.
- Support the preparation of site instructions, contemplated change notices and change order evaluations from a technical standpoint.
- **Site Coordination and Communication**
 - Coordinate with municipal operations, transit and emergency services for lane closures, detours and temporary works.
 - Coordinate with municipal operations for watermain shutdowns and connections.
 - Provide on-site liaison with residents, businesses and facility operators as directed by the owner.
 - Distribute notices to residents and businesses affected by any shutdowns and site works.
 - Help schedule and document site meetings, including action items and follow-up.

Result: **well documented, well-coordinated construction** that supports timely payments, clear communication and defensible decisions.

Integration of Design and Construction Support

A key advantage of working with a focused team like Vatas is the **tight integration between design and field services**.

We integrate design and construction support by:

- **Maintaining Design-Field Continuity**
 - Keeping the same senior engineers involved from planning and design through construction, so context is not lost.
 - Ensuring that any field changes remain consistent with design assumptions, safety factors and regulatory commitments.
- **Closing the Loop on Constructability**
 - Using issues raised during construction (e.g., staging challenges, utility conflicts, detail improvements) to refine standard details, checklists and design practices for future assignments.
 - Incorporating lessons learned into subsequent SPR/SDR work and corridor designs so programs benefit over time.
- **Supporting Close-Out and As-Constructed Documentation**
 - Assisting with review of as constructed information for accuracy and clarity.
 - Providing concise technical summaries where needed for handover to operations and asset management teams.

Result: municipalities receive **projects that not only meet design intent on paper but are also informed by real field experience**, improving the quality of both current and future assignments.

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